



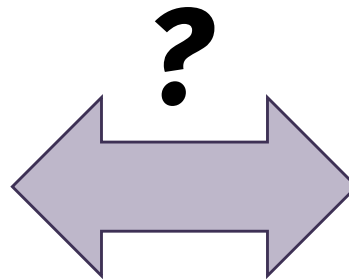
Machine Learning refresh

A Journey through Deep Learning

What is (Machine) Learning

Intelligent Systems

Technologically advanced machines that **perceive** and **respond to** the world around them. Intelligent systems can take many forms (automated vacuums, facial recognition programs, personalized shopping suggestions)



Data Science

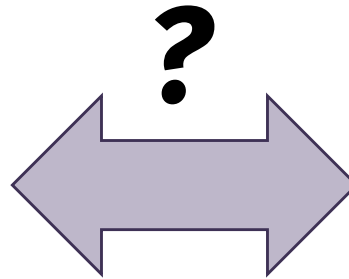
An inter-disciplinary field that uses scientific methods, processes, algorithms and systems to extract knowledge and insights from many structural and unstructured data

BUT ALSO

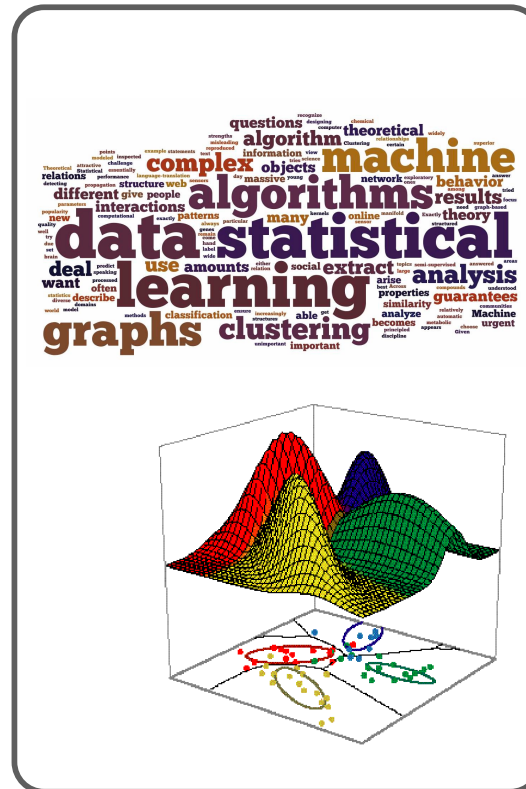
a concept to unify statistics, data analysis and their related methods in order to understand and analyse actual phenomena with data

What is (Machine) Learning

Intelligent Systems



Data Science



Machine Learning

“It is the field of study that gives computers the ability to learn without being explicitly programmed”

- Arthur L. Samuel, AI pioneer, 1959

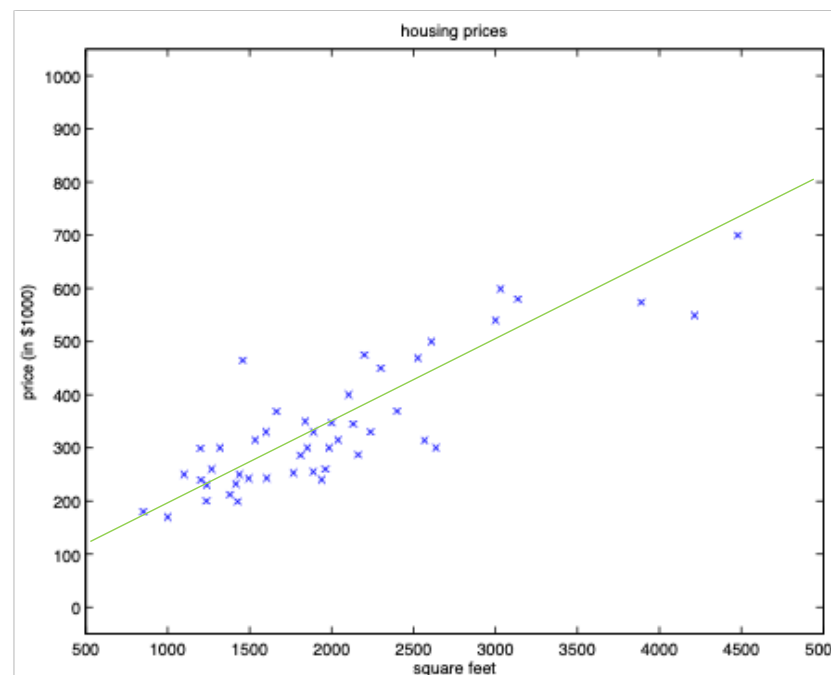
Basic setting: Regression (with an example)

DATA

Living area (feet ²)	Price (1000\$)
2104	400
1600	330
2400	369
1416	232
3000	540
⋮	⋮
2250	?
6000	?

$$(x_1, y_1), \dots, (x_n, y_n)$$

Living area (feet ²)	#bedrooms	Price (1000\$)
2104	3	400
1600	3	330
2400	3	369
1416	2	232
3000	4	540
⋮	⋮	⋮



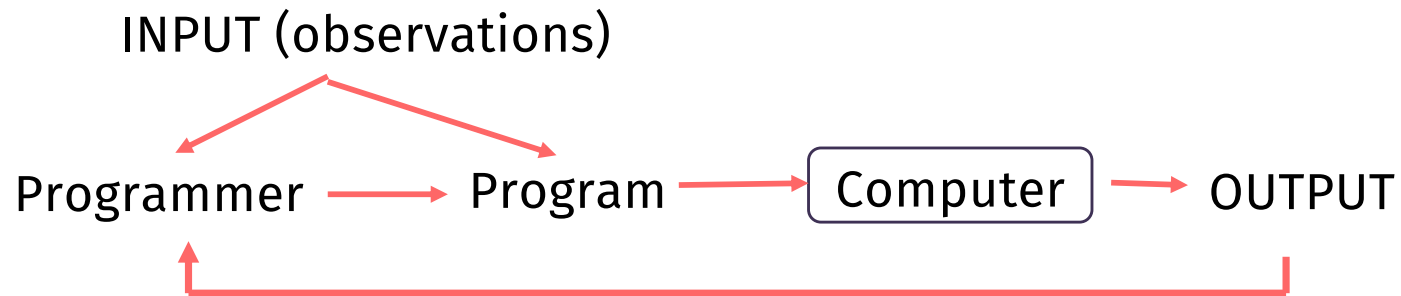
You are looking for a function
such that

$$y_i \simeq f(x_i)$$

And able to generalize to new
data

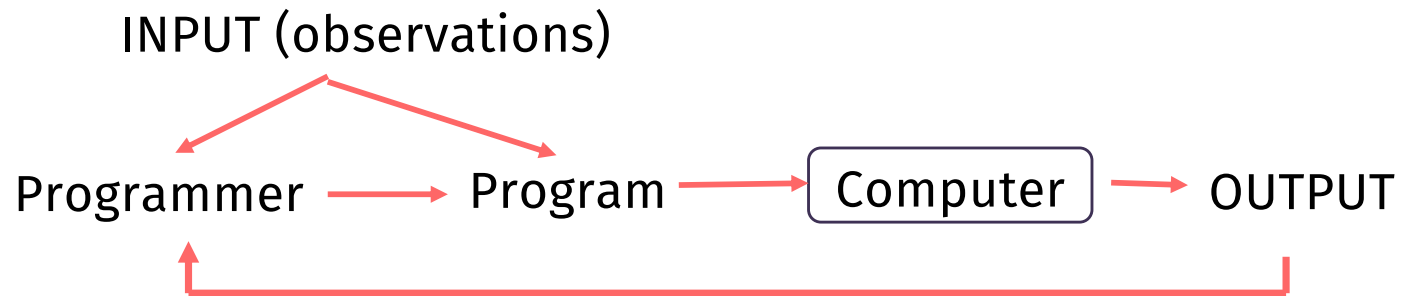
Machine Learning

- Traditional programming paradigm

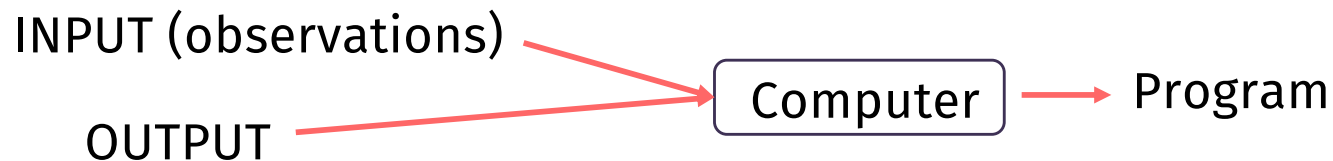


Machine Learning

- Traditional programming paradigm

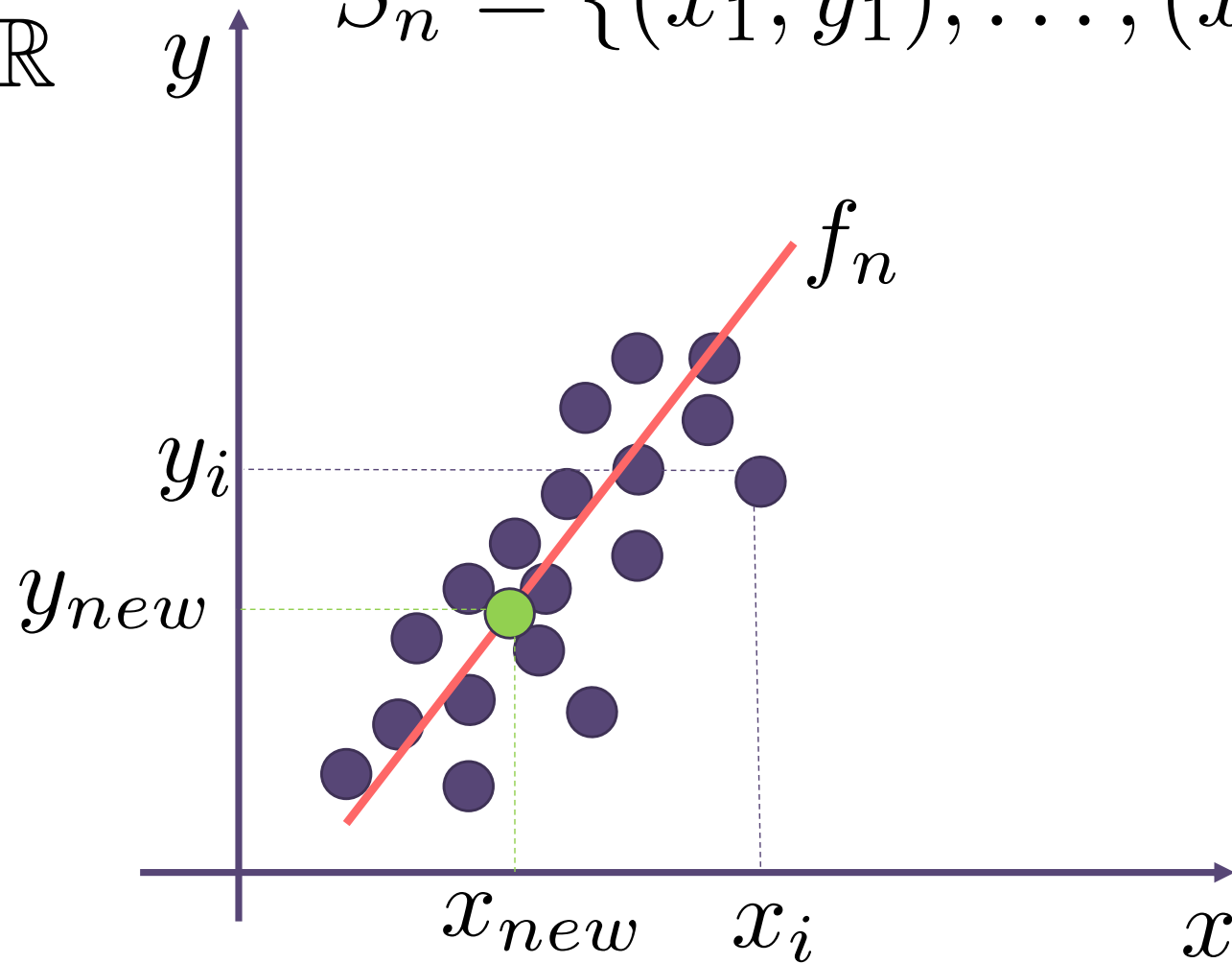


- Machine Learning paradigm



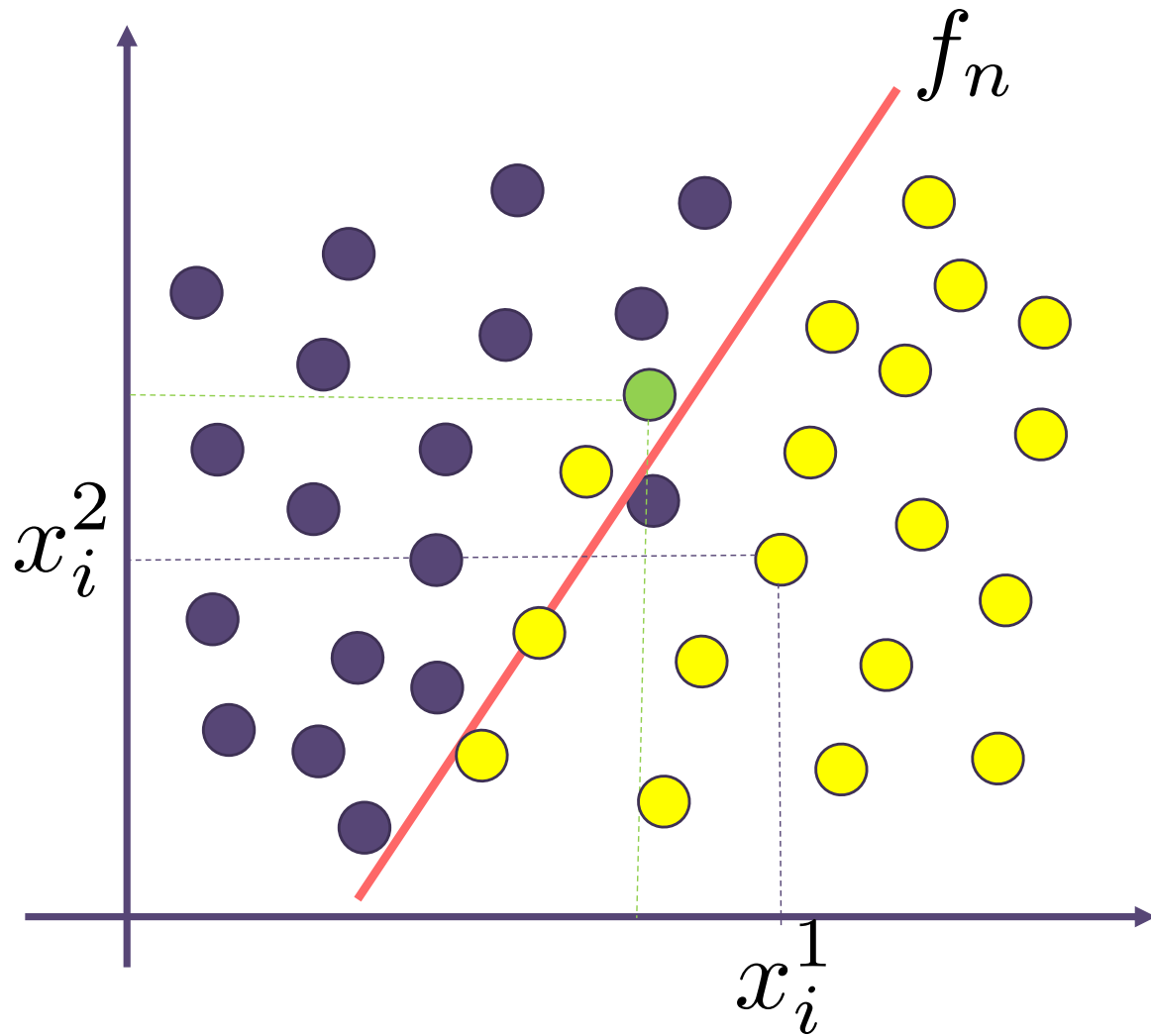
Basic setting: regression

$$Y \subseteq \mathbb{R} \quad S_n = \{(x_1, y_1), \dots, (x_n, y_n)\}$$



Basic setting: classification

$$Y = \{-1, 1\}$$
$$X \subseteq \mathbb{R}^2$$
$$x_i = [x_i^1, x_i^2]$$



$$S_n = \{(x_1, y_1), \dots, (x_n, y_n)\}$$

Supervised Learning

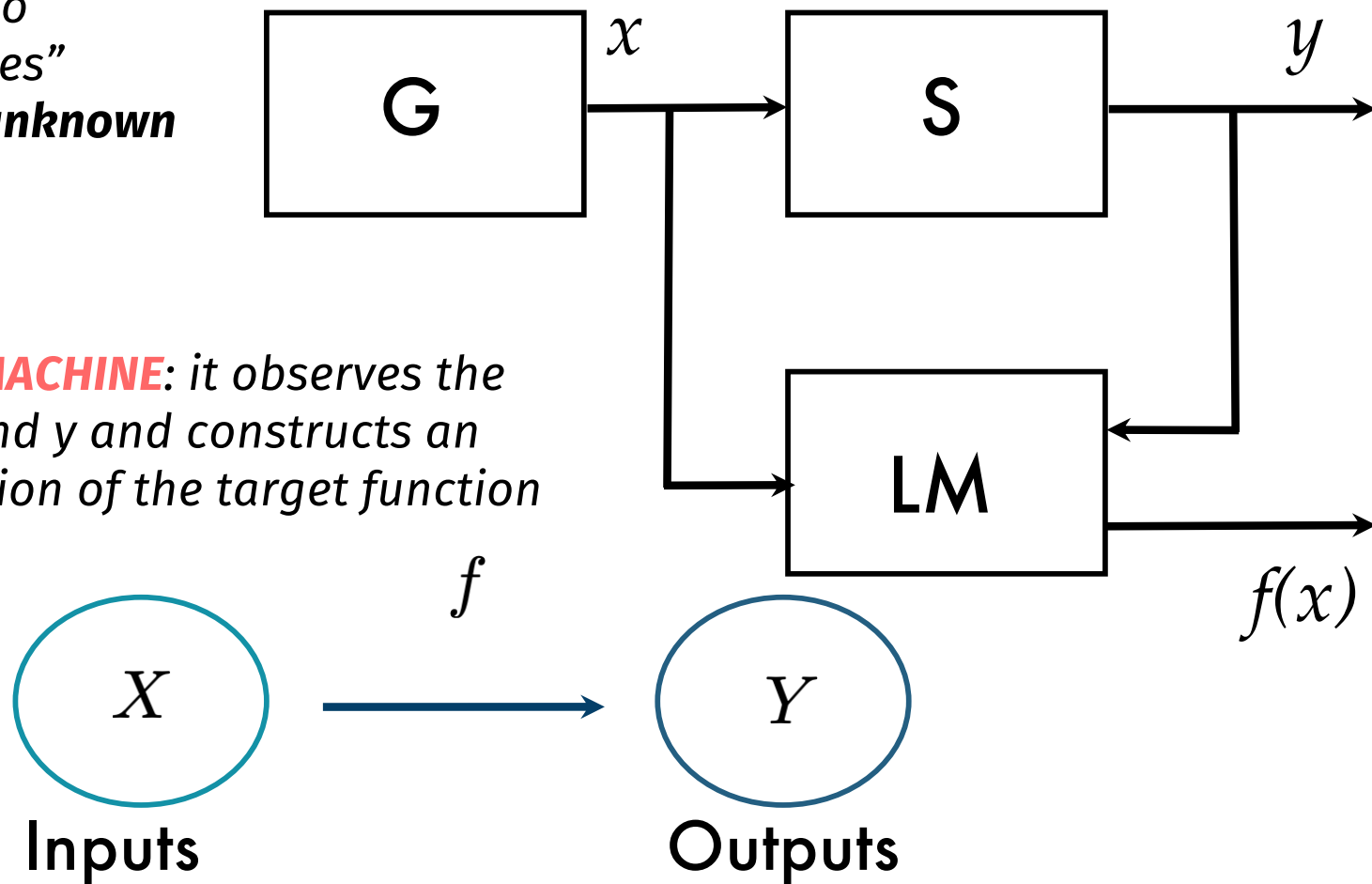
$$(x_1, y_1), \dots, (x_n, y_n)$$

GENERATOR:

generates vectors according to “certain rules” which are **unknown** but **fixed**

SUPERVISOR: transforms the vectors into output values. It is **unknown** but **it exists** and **does not change**

LEARNING MACHINE: it observes the pairs of x and y and constructs an approximation of the target function

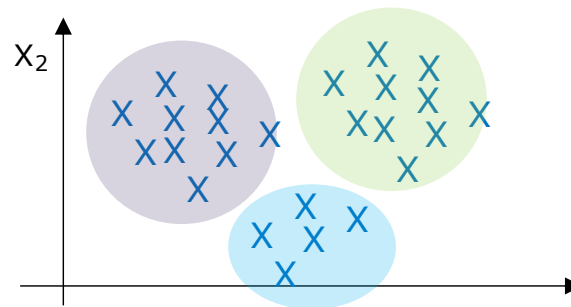


UnSupervised Learning

$$(x_1, \cancel{y_1}), \dots, (x_n, \cancel{y_n})$$

Metaphor: *learning without a teacher*

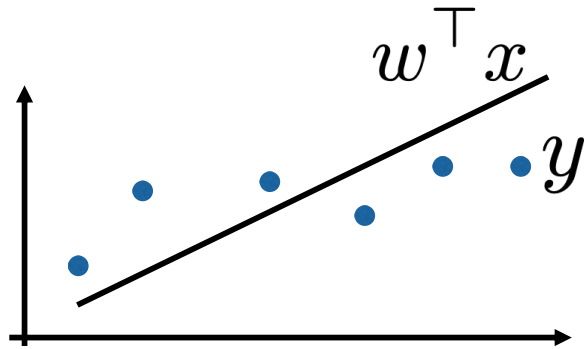
The goal of unsupervised learning is to directly infer **properties** of the data without the help of a supervisor or teacher providing correct answers or degree-of-error for each observation



A quick tour on supervised algorithms

- Linear regression/classification
- Logistic regression/Support Vector Machines
- Neural Networks ...

Linear regression



Model

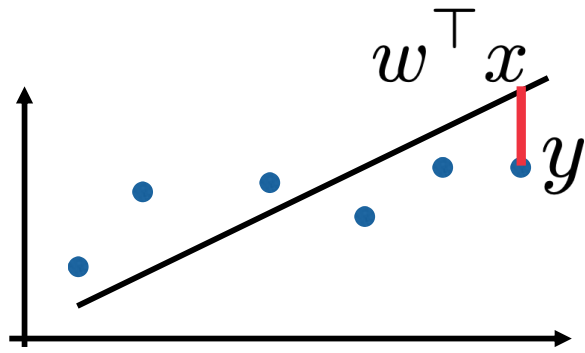
$$f(x) = w^T x = \sum_{j=1}^d w^j x^j$$

Training

$$\min_w \sum_{i=1}^n (y_i - w^T x_i)^2$$

Loss function

Different loss functions = different algorithms



Logistic loss

$$\log(1 + e^{-w^T x})$$

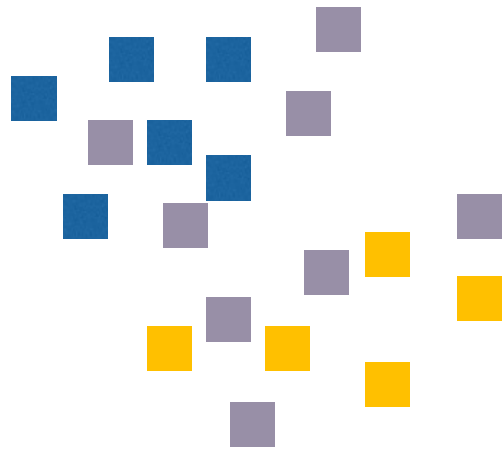
Square loss

$$(y - w^T x)^2$$

Hinge loss (SVM)

$$|1 - w^T x|_+$$

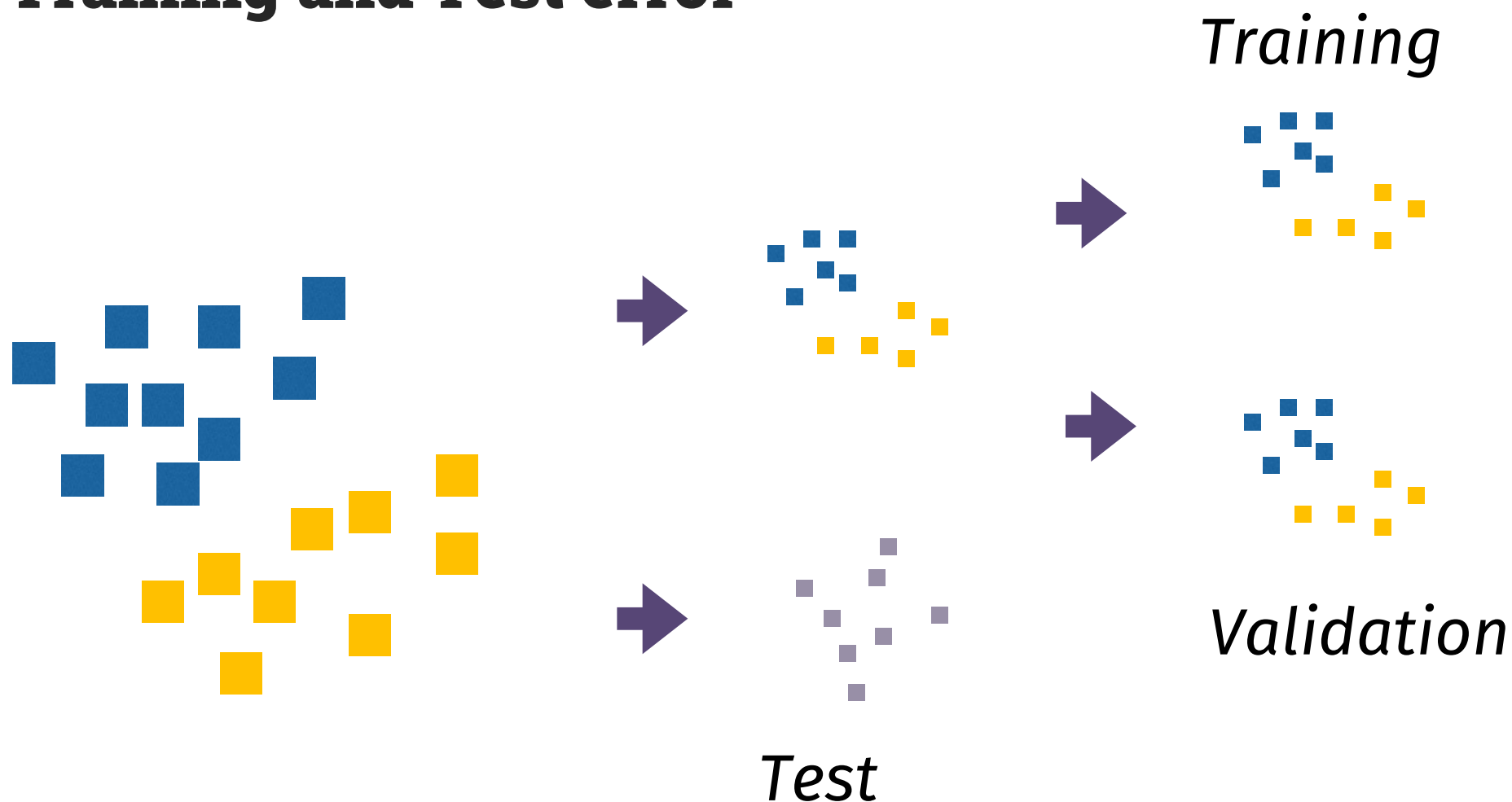
Prediction error



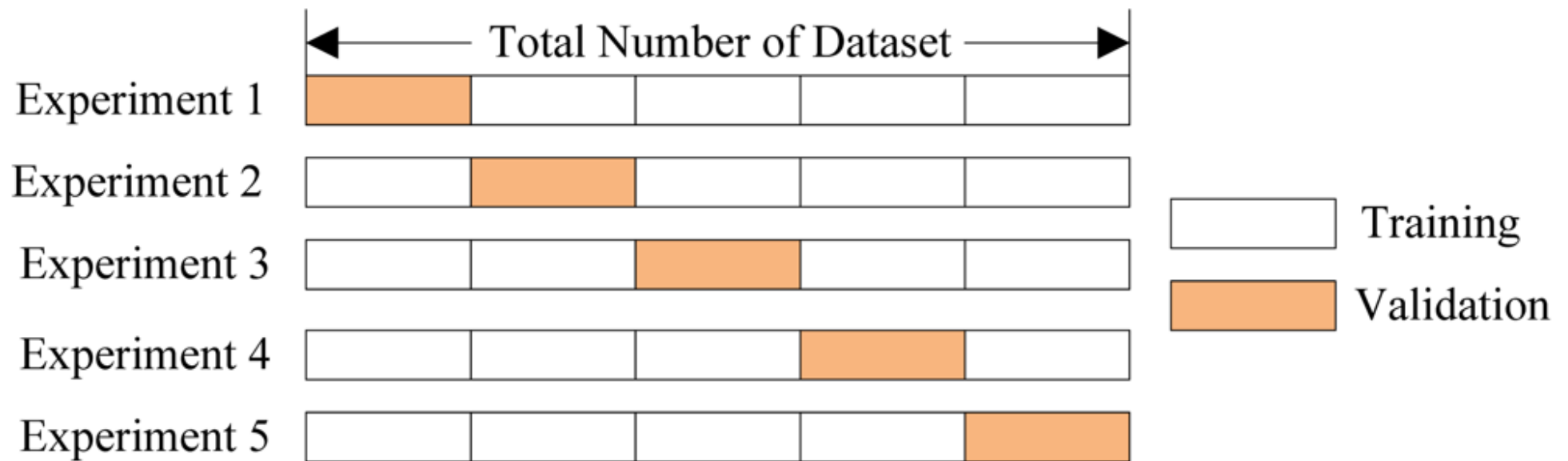
GOAL: given the current data make predictions about future data

...but the future is not known!

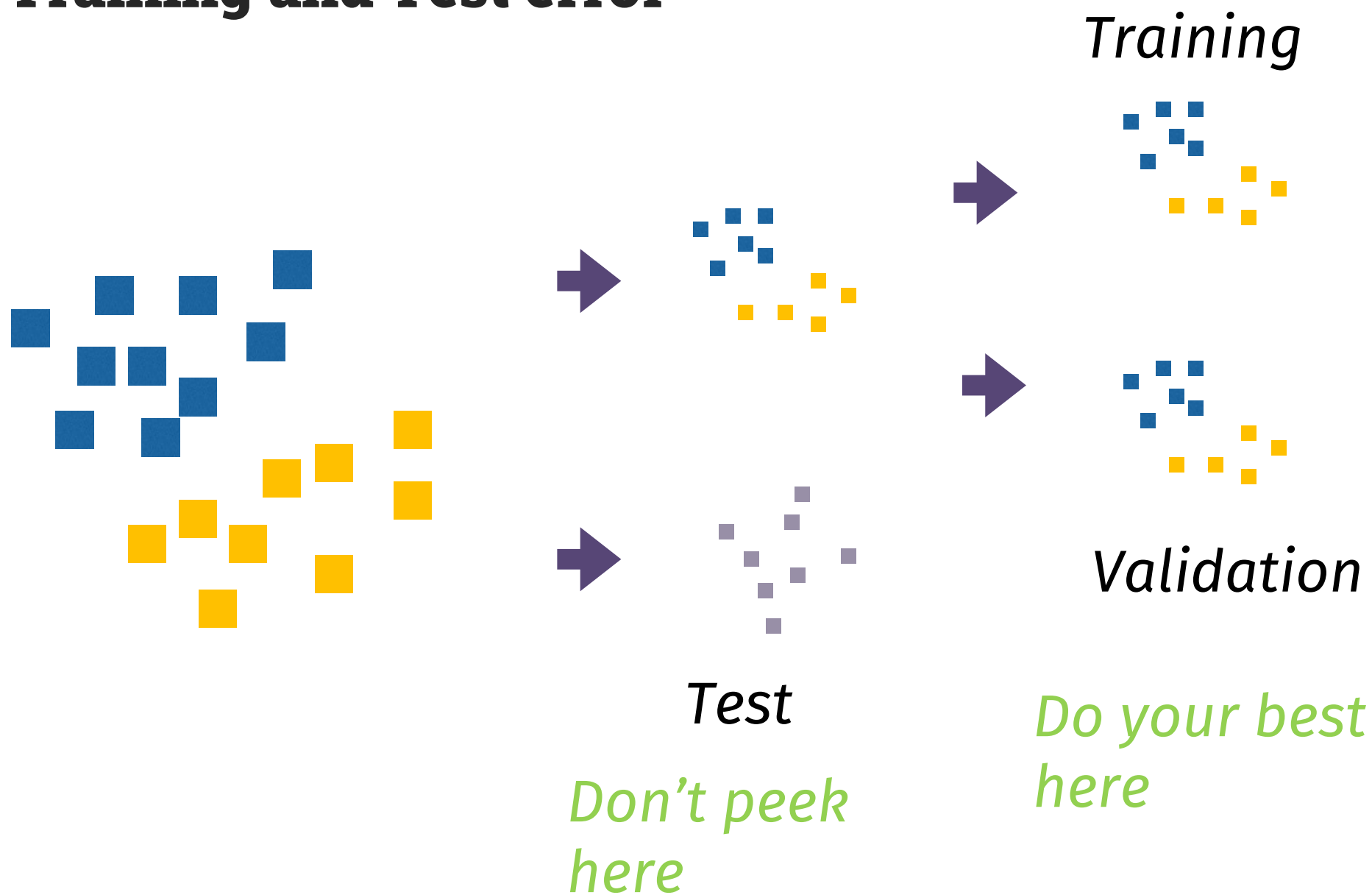
Training and Test error



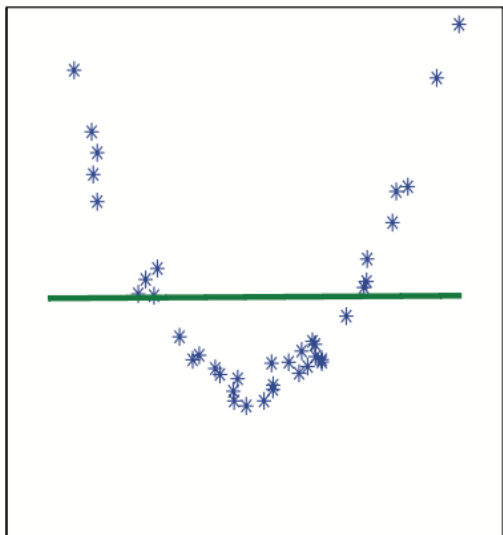
Cross Validation



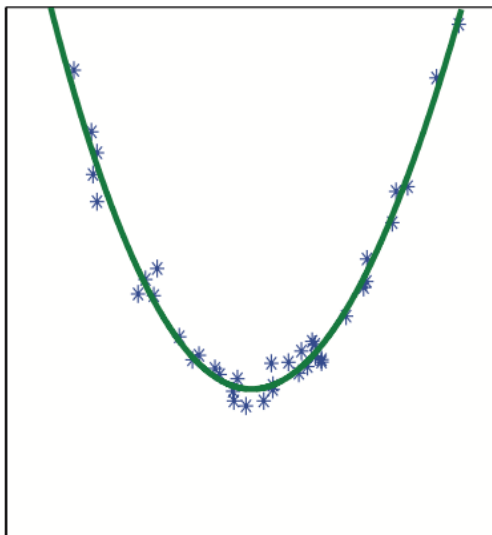
Training and Test error



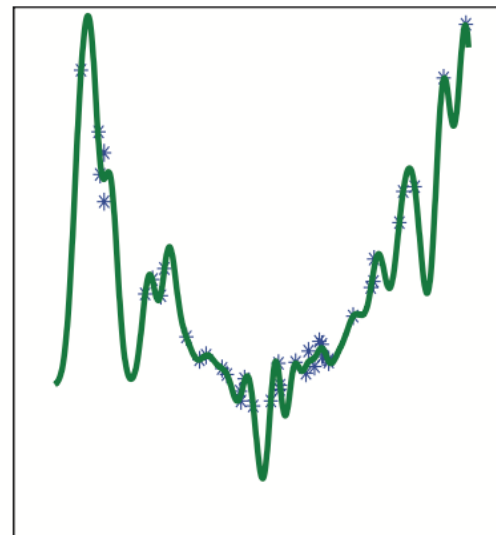
The problem of overfitting



UNDERFITTING
*The model is too smooth
Does not represent the data*



"ideal" fitting



OVERFITTING
*The model is too complex
Does not generalize well*

UniGe

